

Week 2: Linear and Logistic Regression

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What stood out to me in *Regression and Other Stories* was the distinction between using regression for prediction and using it to compare groups. A coefficient can describe how average earnings differ between taller and shorter people, but that alone does not tell us what would happen to someone's earnings if their height changed. Calling a coefficient an "effect" can therefore hide assumptions that the fitted model does not establish. This makes interpreting a regression seem at least as important as fitting it.

I also found it helpful to think of regression as modeling a distribution of outcomes given the inputs. With independent normal errors of equal variance, $y_i | x_i \sim \mathcal{N}(a + bx_i, \sigma^2)$, maximizing the likelihood over a and b gives the same estimates as minimizing squared residuals. This connects a familiar fitting rule to an explicit assumption about the data, although normality is not required simply to fit least squares. Keeping the distribution in mind also clarified regression to the mean: predicted heights can be closer to the average without implying that everyone eventually has the same height, because actual outcomes still vary around those predictions.

Logistic regression extends this perspective to binary outcomes. It models a Bernoulli probability $p(x)$, with $\log[p(x)/(1 - p(x))] = a + bx$, so predicted probabilities stay between zero and one. The coefficient describes a constant difference in log-odds per unit of x ; the corresponding difference in probability depends on the starting value of x . I find the shared likelihood view useful: choosing a distribution for the outcome determines how we fit the model and interpret its predictions. The question I am left with is how to tell when a model predicts reasonably well but gives misleading comparisons because important variables or interactions have been left out.

Readings: Gelman, Hill, and Vehtari, *Regression and Other Stories*; Shalizi, *Advanced Data Analysis from an Elementary Point of View* (logistic regression).